

Problem description

The problem we are aiming to solve is whether it is possible to predict how sensitive a cancer cell line is to a drug via its biological and pharmacological profile. Across the world, there is a variance in genetic mutations and drug response varies with them leading to the adverse side effects many patients experience. Scientists across the globe study the conserved sequences among patients to help them understand certain cancers more and use the mutations within the sequences to create medications, which raises the question around how a specific genomic factor would have a role in the efficacy of Cancer Therapy drugs.

Through this project, the models being used to analyze the large cancer drug sensitivity dataset will be ElasticNet and LightGBM. These models utilize methods that will be useful in examining the relationship between genomic profiles and drug efficacy. The ElasticNet model aims to determine whether drug sensitivity can be predicted based on biological and pharmacological characteristics, using the Lasso and Ridge methods. The second model, LightGBM, is a decision-based tree of algorithms used for ranking, classification, and others. Ultimately, these models will help reach the end goal of reducing adverse drug effects and improve the treatment experience for patients undergoing cancer therapy.

Project scope

The scope of this project is to create a high-performing data science based solution that can predict cancer drug sensitivity based on genetic features. The application of exploratory data analysis and machine learning techniques the project will find identify important relationships. This insight will allow healthcare professionals to provide more information on how individual patients may respond to specific cancer drugs and overall improve treatment effectiveness while reducing adverse side effects provided from inaccurate drug dosage.

The primary modelling approaches taken will be ElasticNet regression and LightGBM. ElasticNet combines L1 and L2 regularization to handle multicollinearity and perform feature selection making it a strong choice for the genomic data. LightGBM, which is a gradient boosting approach, is efficient and uses techniques such as GOSS and EFB to improve the speed without sacrificing accuracy. The model will then be evaluated based on R-squared and RMSE values showing the two model strong predictive abilities.

Finally, the primary deliverable of this project is an analysis of drug efficacy patterns and the identification of possible relational features, such as genes that may influence drug response. This analysis could support the development of an interactive dashboard where healthcare professionals and researchers input relevant data to generate possible IC50 predictions. This is targeted toward healthcare professionals and researchers to communicate our seemingly complex models into readable insight. The dashboard would allow the user to explore various predicted drug responses for various genetic profiles, view key features that influence treatment results, and compare different effectiveness of various treatment plans. Overall, the project aims to produce a strong analysis of the relationships uncovered between genomic variations and drug response.

Metrics

This project's goal is to address the drug response prediction in a supervised learning setting. The Target variable is LN_IC50 (which is the natural log of the drug concentration needed to reduce 50% of the tumor/cancer). The more negative the LN_IC50, the more sensitive the drug is to that specific cell line or cancer, and the more positive it is, the more resistant it gets. Since the target is continuous, it has a range of -8.75 to 13.82 across the main dataset, with no missing values requiring major preprocessing.

3 evaluation metrics would be used to evaluate these models. RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and R-squared. RMSE would measure the average prediction error in LN_IC50, while also penalizing large errors more than small ones due to squaring the differences. This is a good metric considering that the LN_IC50's extreme sensitivity or resistance values are the most significant. MAE is easier to interpret because it shows the average difference in the same units as LN_IC50, without squaring penalties. The R-squared score captures the variance in the drug responses that the model explains, where a value near 1 would indicate a good fit, and a value near 0 would indicate a bad fit. A baseline model that would predict the mean of LN_IC50 across all training data could achieve an R-squared score of 0.0, while the RMSE and MAE would be determined by the dataset's distribution. ElasticNet regression serves as the modelling baseline, a linear model that establishes the floor performance that tree-based models must meaningfully exceed to justify their complexity.

Since the linear models are expected to struggle with the high-dimensional and sparse nature of genomic features based on the preprocessing approach, a tree-based model, such as lightGBM and CatBoost, might have better evaluation performance. The target threshold for a model is considered good when the R-squared is above 0.70 under default parameters, as any fine-tuning on a model that has an R-squared above 0.70 could make the prediction better. This threshold is evaluated against the GDSC dataset, the same resource used in the original screening done by Garnett et al. (2012), which established that genomic features are systematically associated with drug sensitivity across cancer cell lines[1].

Architecture

The project uses the GDSC dataset to predict drug sensitivity (LN_IC50) in a supervised regression setting. The data is processed in Python using Pandas, NumPy, and Scikit-learn. Categorical features are encoded, numeric features are scaled, and genomic mutation data is transformed into a structured format suitable for modeling.

To prevent data leakage, the dataset is split using GroupShuffleSplit by COSMIC_ID, ensuring that the same cell line does not appear in both the training and testing sets. This helps maintain a more realistic evaluation of model performance.

Modeling is performed using ElasticNet, LightGBM, and CatBoost, allowing the project to capture both linear relationships and more complex non-linear interactions within the data. ElasticNet serves as a regularized linear baseline, while the tree-based models handle high-dimensional and sparse feature spaces more effectively.

Model performance is evaluated using RMSE, MAE, and R^2 , which measure prediction error and how well the model explains variation in drug response.

The overall systems goal is to process and analyze cancer and drug-related features and receive predicted drug sensitivity, along with supporting visual insights.

Plan

Phase 1: Data acquisition and understanding

- Confirm dataset structure, target variable and feature types
- Confirm data is workable and note necessary preprocessing steps
- Consolidate data from various sources to one file

Phase 2: Preprocessing and feature engineering

- Address missing values, scale and normalize numeric features, encode categorical features, and split the data into training/validation/test sets

- Assess range of available data- different cancer types/available treatments
- Determine which data should be excluded altogether

Phase 3: Exploratory Data Analysis

- Establish data statistics, trends, primary model
- Visualize trends/ vector relationships via heatmap, histogram

Phase 4: Model development and baseline results

- Preliminary model for association between pharmacogenomic markers and treatment as well as additional information provided

Phase 5: Evaluation, Interpretation, and Variant Associations

- Evaluate model, improve accuracy if possible
- Analyze the output and interpret the drug effectiveness

Phase 6: Prototype Output and final deliverables (if possible)

- Develop interactive dashboard with relationship features

Personnel

The team consists of six collaborators working on the Cancer Pharmacogenomics Prediction Project. All team members will participate in the data preprocessing phase, which includes cleaning the datasets, handling missing values, merging supplementary datasets, and preparing features for modeling. This shared effort ensures that all members understand the data's structure and limitations of the data before the modeling phase begins.

After preprocessing, team responsibilities will be divided according to areas of focus and individual strengths. Allan and Mohamad will lead the model development phase, implementing supervised learning algorithms and building models to predict drug sensitivity using the prepared dataset. Hiu and Em will focus on model evaluation and validation. Their responsibilities include analyzing model performance using evaluation metrics such as RMSE, MAE, and R-squared, comparing different models, and determining which approach provides the most reliable predictions. Grace and Julie will lead the analysis and narrative development portion of the project. They will draft the written explanation connecting the exploratory data analysis (EDA), preprocessing decisions, and modeling results to the core research question: "Can we predict how sensitive a cancer cell line will be to a drug, given its biological and pharmacological profile?"

Although specific responsibilities are assigned to promote efficiency, all team members will collaborate during discussions, review intermediate results, and contribute to the final documentation and presentation of the project findings.

Communication

The group's primary forms of communication will be text messaging, email, and ClickUp. Text messaging and email will be used for quick updates, scheduling meetings, and addressing questions that arise during the project. The team will schedule full-group meetings at times when all members are available to discuss project progress, address challenges, and plan upcoming tasks. To keep the project organized and transparent, the team will use ClickUp as the primary work management platform. ClickUp will be used to track tasks, deadlines, and responsibilities through visual task boards, timelines, and status updates. Features such as task assignments, progress tracking, and shared project views will centralize project information and ensure all team members are informed about ongoing work. In addition, the team will use GitHub as a shared repository for version control and collaboration. All team members will upload their code, notebooks, and relevant project files to GitHub to ensure consistent version tracking and accessibility.

References

- 1) Garnett, Mathew J., et al. "Systematic identification of genomic markers of drug sensitivity in cancer cells." *Nature*, vol. 483, no. 7391, Mar. 2012, pp. 570–575, <https://doi.org/10.1038/nature11005>.